

DEALFLOW360

The Intelligent Deal Co-Pilot for B2B Sales

Product Vision · System Design · Competitive Differentiation

A self-governing deal engine that acts, not just reacts — enforcing pricing discipline, routing approvals intelligently, splitting fulfillment in real time, and co-piloting every deal from first quote to closed won.

Problem Statement: DealFlow360 — Intelligent Sales Operations Platform

Team: 3 Members | Backend · Frontend · AI/Logic

Document Type: Product Design + System Architecture

Approach: Glass Box Logic + Stateful AI Co-Pilot + SaaS-Grade UX

01 | The Product's Reason to Exist

THE CORE BELIEF

B2B deals don't fail because reps are bad at selling.

They fail because **information is trapped** — in the rep's head, in email threads, in approval queues, in static PDFs.

DealFlow360 makes the deal itself intelligent, so information flows automatically and the system carries the cognitive load while the rep just closes.

What DealFlow360 Replaces

The Old Way	Why It Fails
■ Salesforce / HubSpot	Too complex, built for CRM not deal execution
■ Excel quote sheets	No live logic, no approval routing, no collaboration
■ Email negotiation	Slow, untracked, no audit trail, customer outside the deal
■ Manual manager review	Manager has no context, every deal looks the same
■ DealFlow360	Deal-aware, self-routing, customer-inside, co-piloted

Four Product Principles

The System Has Opinions	It doesn't wait to be asked. It watches deal state, reasons about context, and surfaces the right action at the right moment with a draft ready.
Nothing Is a Black Box	Every decision — routing, scoring, splitting, flagging — is visible, interactive, and explainable. Reps trust what they understand.
Less Clicks, More Closes	Every screen reduces time between 'quote built' and 'deal signed'. If a screen doesn't serve that goal, it doesn't ship.
The Customer Is Inside the Deal	Negotiation happens inside the platform. The customer portal is a first-class experience, not an afterthought or a relabelled internal screen.

02 | Requirements Coverage + What We Add On Top

The spec defines a floor — 8 modules every team will build. Below is what the problem statement requires (■), what most teams will fake or skip (■■), and what DealFlow360 adds beyond the spec (■).

Feature / Requirement	Status	Our Implementation
Approval routing (auto, no rep action)	■ Required	Blended risk score per line + category ceiling, not just order-level %
Blended discount risk score	■ Required	Per-line overage + cross-line accumulation — correct math, not faked
Upsell / cross-sell panel	■ Optional	Co-purchase history + live margin delta, dismissable, updates total instantly
Warehouse split + backorder	■ Required	Cost/shipment optimization math shown openly with manual override
Hybrid billing (one-time + recurring)	■ Required	Proration rules, billing schedule, credit note trigger on cancellation
Customer portal — separate, restricted	■ Required	Separate auth, separate visual identity, counter triggers re-approval
Deal health + anomaly dashboard	■ Required	Stall detection, discount anomaly vs rep's historical avg, slippage alerts
Full audit trail on all approvals	■ Required	Every action: user + timestamp + reason + state change — visual timeline
Glass Box Risk Breakdown UI	■ Extra	Click any risk score → per-line breakdown + one-click fix suggestion
Deal Co-Pilot Feed (state-triggered)	■ Extra	AI narrates deal events as they happen — not a chatbot, triggered by data
Counter-negotiation AI draft	■ Extra	When customer counters, co-pilot drafts rep's response before rep opens it
Stall escalation with reasoning	■ Extra	System detects stall, writes escalation context, rep clicks send or edit
Persistent co-pilot memory	■ Extra	Feed shows what happened while rep was offline — system worked overnight
SaaS-grade design system	■ Extra	Consistent tokens, motion design, empty states, error states, loading states

■ Rows 1–8 = Spec requirements | ■ Rows 9–14 = What we build beyond the spec

03 | The Deal Co-Pilot — Core Differentiator

DEFINITION

The co-pilot is not a chatbot. It is a state-triggered reasoning engine. It watches deal data change, packages context, reasons about what it means, and writes actionable cards into a persistent Deal Intelligence Feed. The rep never has to ask it anything.

What Triggers the Co-Pilot

Trigger (State Change)	Co-Pilot Action
Quote line added / modified	Recalculates risk, updates blended score narration
Discount crosses category ceiling	Explains which line, by how much, offers fix
Approval submitted	Confirms new path, estimates time to approval
Customer counter received	Drafts rep's response before rep opens notification
Deal inactive > X hours	Detects stall, writes escalation with context
Warehouse stock drops	Flags fulfillment risk on open quotes using that stock
Subscription quantity changed	Calculates proration, shows credit note if applicable
Finance approver offline	Suggests alternative escalation path

Feed Card Examples — The Co-Pilot's Voice

Cards are concise, specific, and always end with an action. The system never just reports — it recommends.

■ Risk Escalation · 2 min ago

Setup Service line crossed its category ceiling by 8 pts. Quote now requires Finance approval (was Manager only).
→ Reduce Setup Service to 13% to drop back to Manager only.

[\[Apply Fix\]](#) [\[Keep & Escalate\]](#)

■ Upsell Opportunity · just now

3 of last 4 Laptop Pro buyers added Extended Warranty within 30 days. Adding it here lifts order margin by 2.1% with no ceiling impact.

[\[Add to Quote\]](#) [\[Dismiss\]](#)

■■ Deal Stalled · 31 hrs ago

Acme Corp quotation has been inactive for 31 hours. Finance approver Rohan last logged in Tuesday. Close date is in 3 days. Escalation draft prepared for Sales Director.

[\[Review Draft\]](#) [\[Snooze 24h\]](#)

■ Customer Countered · just now

Customer requested 22% on Setup Services (currently 18%). Their 3-year history shows ■21L in purchases.
Recommended counter: 14% with 2-year subscription lock-in. Draft ready.

[\[Review Draft\]](#) [\[Send Now\]](#) [\[Decline\]](#)

AI Model Strategy

Core business logic (blended risk, routing, warehouse split) is pure deterministic math — zero AI, zero latency, zero hallucination risk. AI is used surgically only where judgment or natural language adds value.

Task	Model	Rationale
Co-pilot feed narration	Claude (Anthropic)	Best contextual reasoning, professional tone
Counter-negotiation draft	Claude (Anthropic)	Nuanced, business-appropriate language
Upsell reasoning	Gemini Pro	Fast, structured suggestion generation
Stall detection summary	GPT / Gemini	Simple summarization — save Claude tokens
Blended risk narration	Pure Math — No AI	Always correct, instant, no token cost
Approval routing decision	Pure Math — No AI	Deterministic — must never hallucinate
Warehouse split optimization	Pure Math — No AI	Algorithm-based cost/shipment minimization

04 | System Architecture

High-Level Architecture

PRESENTATION LAYER	Rep Workspace (Next.js) · Customer Portal (Next.js — separate route) · Admin Config Panel
CO-PILOT ENGINE	Context Packager → Trigger System → LLM Reasoning (Claude / Gemini / GPT) → Feed Generator
CORE LOGIC LAYER	Blended Risk Engine · Approval Router · Warehouse Splitter · Billing Engine · Anomaly Detector · Audit Logger
API LAYER	FastAPI (Python) · REST /api/v1/ · JWT Auth · Role-based middleware · Redis pub/sub for feed
DATA LAYER	PostgreSQL — Deals · Products · Customers · Warehouses · Subscriptions · Approvals · Audit Log

Core Data Flow — Quote to Co-Pilot Feed

1	Rep adds/modifies quote line Quote Builder UI sends PATCH to /api/v1/quotes/{id}/lines
2	Backend recalculates Blended Risk Engine re-scores all lines + accumulation check
3	State change detected Trigger System compares new state vs previous state snapshot
4	Context packaged Deal metadata + line deltas + rep history → structured prompt
5	LLM called async Claude/Gemini generates feed card text (non-blocking)
6	Feed card written Card stored in DB, pushed via Redis pub/sub to frontend
7	UI updates live Deal Intelligence Feed shows new card, rep sees it instantly
8	Fallback on failure If LLM fails, deterministic template fills card — never blank

05 | Key Screen Designs — The Glass Box UX

Quote Workspace — Three-Panel Layout

The rep's primary working environment. Three persistent panels: product picker (left), live order builder with margin bar (center), Deal Intelligence Feed / co-pilot (right). A persistent bottom bar shows: Blended Risk Score (clickable) · Live Margin % · Estimated Deal Value · Approval path required. Every value updates on every keystroke — no save required to see impact.

- Left: Product picker with category tabs + margin indicator per product
 - Center: Order lines, line-level discount, quantity $\pm$, running totals
 - Right: Co-pilot feed with timestamped action cards
 - Bottom bar: Risk score (clickable) · Margin % · Total · Approval path
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Risk Breakdown — The Glass Box Moment

Click the risk score anywhere in the product to open this drawer. This is the single most differentiated screen — no other team will show their math. Judges see the system is correct, not just convincing.

- Per-line table: product · discount given · ceiling · overage · status
 - Progress bars showing how close each line is to its ceiling
 - Blended score calculation shown step by step
 - Approval path derived from score with explanation
 - Actionable fix: 'Reduce X to Y% → score drops to Z → Manager only'
 - One-click [Apply Fix] button — no manual editing required
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Deal Intelligence Feed — The Co-Pilot Panel

Persistent right panel in the workspace. Scrollable card feed. Each card has a colored left border (red=risk, amber=warning, green=opportunity, blue=info), timestamp, one-paragraph reasoning, and 2–3 action buttons. Feed persists across sessions — rep sees what happened while offline.

- State-triggered — cards appear automatically on data changes
 - Color-coded severity: red · amber · green · blue
 - Every card ends with an action button — never just informational
 - Persistent across sessions — system worked while rep was away
 - Fallback cards use deterministic templates if LLM is unavailable
-

Customer Portal — A Different World

Completely separate route (/portal), separate visual identity — cleaner, calmer, customer-facing. Customer authenticates via magic link. Sees their quotation with current status, can comment per line, propose a counter-discount, or confirm final terms. Submitting a counter automatically triggers re-approval if thresholds are breached.

- Separate auth (magic link), separate domain/route, separate visual design
 - Quotation status timeline: Sent → Under Negotiation → Confirmed
 - Line-level comment and question tool
 - Counter-discount field with message — not just a number
 - One-click confirm locks terms and moves to fulfillment
 - Auto re-approval trigger if counter exceeds thresholds
-

Deal Health War Room — Operational Dashboard

Manager view. Not a metrics dump — an operational snapshot. Four KPI cards: Stalled Deals · Discount Anomalies · Fulfillment Risks · Pending Approvals. Each card is clickable and opens the deal directly with co-pilot context pre-loaded. System can auto-send escalation nudges.

- KPI cards: stalled · anomalies · fulfillment risk · pending approvals
 - Each alert card: deal name · value · days stalled · co-pilot reasoning
 - Click any card → opens deal with full context loaded
 - Anomaly: compares current discount vs rep's 90-day historical average
 - Auto-escalation: system sends nudge if rep doesn't act within threshold
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06 | Core Business Logic — Implemented, Not Faked

The spec explicitly requires: 'Core business rules must be implemented in application logic, not hardcoded or faked for the demo.' Below is exactly how each rule is implemented.

Blended Discount Risk Score — Algorithm

Step	Name	Logic
Step 1	Per-line overage	For each line: $\text{overage}_i = \max(0, \text{discount_given} - \text{category_ceiling})$
Step 2	Weighted overage	$\text{weighted}_i = \text{overage}_i \times (\text{line_value} / \text{order_total})$
Step 3	Blended score	$\text{blended_score} = \sum \text{weighted}_i \times 100$
Step 4	Routing decision	score 0–25 → None 25–50 → Manager 50+ → Manager + Finance
Step 5	Worst-line check	Even if blended < 25, any single line overage > 15pts → escalate

Warehouse Split Algorithm

#	Step	Logic
1	Stock query	Query live stock per warehouse for all order lines
2	Coverage matrix	Build matrix: warehouse × product × available_qty
3	Minimize shipments	Greedy: fulfill as many lines as possible from fewest warehouses
4	Cost weighting	When tie: prefer warehouse with lower configured shipping weight
5	Backorder flag	Lines with zero coverage → backorder flag + ETA from replenishment rule
6	Manual override	Rep can drag quantities between warehouses; cost/shipment recalculates live

Hybrid Billing Logic

Scenario	System Behaviour
One-time lines	Invoice generated immediately on order confirmation
Recurring lines	Subscription record created with plan (monthly/quarterly/yearly)
Proration — start	First invoice = $(\text{days remaining in cycle} / \text{total days}) \times \text{monthly rate}$
Proration — quantity	Mid-cycle quantity change: credit note for unused days + new invoice
Cancellation	Credit note auto-generated for unused portion of current billing cycle

Scenario	System Behaviour
Mixed order display	One-time and recurring shown in separate sections, single order record

07 | Team Structure & Parallel Execution

Role	Codename	Owns
Person 1 — Backend	The Brain	Data model · Business logic engines · API layer · Auth · Seed data
Person 2 — Frontend	The Face	Design system · All screens · Glass box UX · Customer portal · Dashboard
Person 3 — AI/Logic	The Voice	Context packager · Trigger system · LLM integration · Feed card templates · Fallback logic

24-Hour Parallel Execution Plan

Person 1 — Backend

0–3h	DB schema + seed data + JWT auth + role middleware
3–6h	Products, price lists, discount tier APIs
6–10h	Blended risk engine + approval routing (pure math)
10–14h	Warehouse split algorithm + hybrid billing logic
14–18h	Customer portal backend + re-approval trigger
18–22h	Deal health queries + anomaly detection
22–24h	Bug fixes + demo data tuning

Person 2 — Frontend

0–2h	Design system: color tokens, typography, base components
2–6h	Quote workspace: three-panel layout + live margin bar
6–10h	Risk breakdown glass box drawer + approval timeline
10–14h	Deal Intelligence Feed card component system
14–18h	Customer portal (separate route, separate visual identity)
18–21h	Deal health war room dashboard
21–24h	Polish, transitions, empty states, demo rehearsal

Person 3 — AI/Logic

0–2h	Schema review, understand state model for context packaging
2–5h	Context packager: deal state → structured LLM prompt
5–9h	Trigger system: state change detection + async dispatch
9–13h	Counter-negotiation draft engine (Claude)

13–17h	Stall detection + escalation draft (Gemini/GPT)
17–20h	Upsell reasoning layer + feed card formatting
20–24h	Prompt tuning + fallback templates + demo rehearsal

08 | 5-Minute Demo Script

DEMO PRINCIPLE

Every beat has one 'judge moment' — the specific second where a judge reacts. Build to that moment. Everything else is setup.

Minute
1

The Opening — Set Expectations

Open the Deal Health War Room. Three stalled deals. Two anomalies. One fulfillment risk. All flagged automatically — no rep action required. Say: 'Most platforms wait for the rep to notice. DealFlow360 already did.'

■ Judge moment: Seeing the system flagged issues overnight, proactively.

Minute
2

Building a Quote — Co-Pilot Activates

Open a new quote for Acme Corp. Add Laptop Pro × 3. Add Setup Service with 18% discount. The risk bar jumps from green to red. A co-pilot card appears in under 2 seconds: 'Setup Service crossed ceiling by 8pts. Finance now required. Apply 13% fix?'

■ Judge moment: The system reacted before the rep did anything.

Minute
3

The Glass Box — Show the Math

Click the risk score. The breakdown drawer opens. Per-line table: Laptop clear, Setup Service red +8pts. Click [Apply Fix]. Discount updates. Score drops. Approval path changes to Manager only. Say: 'Every other team shows a number. We show the math.'

■ Judge moment: Seeing the logic openly — line by line, with a one-click fix.

Minute
4

Customer Counter — Co-Pilot Responds First

Open customer portal in a second browser tab (different visual design). Submit a counter: 22% on Setup Services. Switch back to rep view. Before the rep opens the notification, a co-pilot card is already there: 'Customer wants 22%. Recommended counter: 14% with 2yr lock-in. Draft ready.'

■ Judge moment: The system drafted the response before the rep clicked anything.

Minute
5

The Close — What This Really Is

Approve the quote, confirm fulfillment split (shown with reasoning), confirm order. Invoice generated. Subscription billing schedule visible. Say: 'We didn't add AI to a sales tool. We built a deal co-pilot and gave it a sales platform to operate. The rep approved three things. The system did the rest.'

■ Judge moment: The rep barely acted. The deal closed itself.

09 | Why This Wins Among 900 Teams

How Most Teams Will Build DealFlow360

What Judges Test	What Most Teams Ship
Risk score	A number or colored badge in the UI
Approval routing	Hardcoded 'if discount > X then show approval screen'
Warehouse split	Static split or random distribution across warehouses
AI integration	A chat sidebar powered by one LLM API call
Customer portal	Internal screen with a different label and restricted nav
Deal health	A dashboard with static charts and hardcoded demo data
Billing	One invoice generated, subscription mocked or skipped

How DealFlow360 Answers the Same Tests

What Judges Test	What DealFlow360 Ships
Risk score	Per-line breakdown, weighted accumulation, shown openly with fix suggestion
Approval routing	Algorithm: blended score + worst-line check → routing decision with explanation
Warehouse split	Greedy minimization algorithm, cost weighting, live recalculation on override
AI integration	State-triggered multi-model layer: Claude + Gemini + GPT, surgically applied
Customer portal	Separate route, separate auth, separate visual identity, real re-approval trigger
Deal health	Live data, anomaly vs rep's 90-day avg, clickable alerts with co-pilot context
Billing	Proration calculated, credit note triggered, one-time + recurring on same order

The One Sentence That Closes the Pitch

"Every other team told the system what to do and used AI to explain it. We built a system that explains itself — because the logic is correct enough to show openly."

10 | Quick Test Flow Coverage (Spec Section 9)

The spec's 8-step test flow is the judge's checklist. Every step must produce a visible, correct result. Below maps each step to our implementation.

#	Spec Test Step	Pas s	Our Implementation
1	Sign up, configure discount tier, warehouse, subscription plan	■	Admin config panel: tier setup, warehouse CRUD, subscription plan builder
2	Create quote with discount above allowed level	■	Risk bar turns red immediately. Co-pilot card appears within 2 seconds
3	Quote auto-routes for manager approval without rep action	■	Blended risk engine triggers routing automatically on quote save/submit
4	Accept upsell suggestion, confirm total and margin update	■	Co-pilot panel: Add to Quote → order total and margin bar update instantly
5	Approved, stock pulled from correct warehouse, split if needed	■	Warehouse split screen shows algorithm result with cost/shipment reasoning
6	One-time product + recurring subscription billed correctly	■	Billing screen: separate sections, correct proration, billing schedule shown
7	Customer requests bigger discount → quote returns to approval	■	Portal submit → backend re-evaluates → approval flow re-triggered automatically
8	Confirm order, record payment, invoice status updates	■	Payment recording → invoice status update → audit trail entry created

11 | What We'd Build Next

The spec requires a short note on future direction. Below is our product roadmap — framed as a real SaaS, not a hackathon wishlist.

Phase 2 — Intelligence

Rep Performance Coaching	Co-pilot learns each rep's patterns over time. Surfaces when a rep is discounting more than their historical average and suggests why.
Predictive Close Probability	Train on historical won/lost deals to score current pipeline deals by probability of closing — not rep sentiment, actual deal signals.
Competitor Price Intelligence	Rep can log a competitor price mentioned by the customer. Co-pilot suggests a counter-positioning response.

Phase 3 — Scale

Multi-currency & Multi-company	Single platform instance serving multiple legal entities with separate discount structures, approval chains, and reporting.
CRM Integration	Bi-directional sync with Salesforce/HubSpot — deals created in CRM appear in DealFlow360, updates flow back automatically.
ERP Fulfillment Sync	Warehouse split decisions pushed directly to ERP. Stock updates pulled in real time without manual reload.

Phase 4 — Autonomy

Fully Autonomous Escalation	System sends escalation emails and Slack messages on behalf of the rep when a deal stalls and the rep is unresponsive past a threshold.
AI-negotiated Counter Terms	Within pre-configured guardrails, co-pilot can submit a counter-offer to the customer autonomously — rep reviews outcome, not process.